Research Idea Generation Prompt Library

Prompt Library

Ultimate AI Research Toolkit 2026 - Premium Edition

Designed for PhD students, professors, graduate researchers, engineers, and data scientists.

[Idea]	**Research prompts and ideation systems**
[Review]	**Literature synthesis and gap detection**
[Write]	**Paper writing and revision workflows**
[Plan]	**Publication and productivity operations**

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Introduction

This Research Idea Generation library provides 120 original prompts engineered for high-rigor academic and technical workflows. Each prompt includes practical fields to help users move from ideation to reproducible output.

Instructions

- Select prompts by category and difficulty to match project maturity.
- Replace variables in the prompt with your specific discipline, data, and constraints.
- Use Expected Output as the acceptance criteria for quality control.
- Store outputs in versioned folders to preserve decision history.

Prompt Library

1. Research Idea Generation 001: Refine Hypothesis Framing for a method benchmarking campaign

Objective: Build a high-clarity strategy for Hypothesis Framing by improving feasibility, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to feasibility, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Methodology

2. Research Idea Generation 002: Synthesize Problem Selection for a reproducibility-sensitive investigation

Objective: Build a high-clarity strategy for Problem Selection by improving rigor, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to rigor, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Validation

3. Research Idea Generation 003: Translate Feasibility Scan for a cross-disciplinary study

Objective: Build a high-clarity strategy for Feasibility Scan by improving generalizability, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to generalizability, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Writing

4. Research Idea Generation 004: Model Novelty Discovery for an ethics-aware deployment study

Objective: Build a high-clarity strategy for Novelty Discovery by improving reproducibility, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to reproducibility, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Reasoning

5. Research Idea Generation 005: Reframe Hypothesis Framing for a longitudinal dataset

Objective: Build a high-clarity strategy for Hypothesis Framing by improving ethical compliance, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to ethical compliance, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Strategy

6. Research Idea Generation 006: Evaluate Problem Selection for a high-impact conference submission

Objective: Build a high-clarity strategy for Problem Selection by improving cost efficiency, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to cost efficiency, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Execution

7. Research Idea Generation 007: Validate Feasibility Scan for an emerging technology review

Objective: Build a high-clarity strategy for Feasibility Scan by improving computational efficiency, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to computational efficiency, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Optimization

8. Research Idea Generation 008: Map Novelty Discovery for a policy-oriented manuscript

Objective: Build a high-clarity strategy for Novelty Discovery by improving publication readiness, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to publication readiness, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Planning

9. Research Idea Generation 009: Prioritize Hypothesis Framing for an industry-backed experiment

Objective: Build a high-clarity strategy for Hypothesis Framing by improving stakeholder clarity, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to stakeholder clarity, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Methodology

10. Research Idea Generation 010: Forecast Problem Selection for a multi-phase project

Objective: Build a high-clarity strategy for Problem Selection by improving novelty, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to novelty, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Validation

11. Research Idea Generation 011: Operationalize Feasibility Scan for a method benchmarking campaign

Objective: Build a high-clarity strategy for Feasibility Scan by improving feasibility, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to feasibility, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Writing

12. Research Idea Generation 012: Compare Novelty Discovery for a reproducibility-sensitive investigation

Objective: Build a high-clarity strategy for Novelty Discovery by improving rigor, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to rigor, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Reasoning

13. Research Idea Generation 013: Optimize Hypothesis Framing for a cross-disciplinary study

Objective: Build a high-clarity strategy for Hypothesis Framing by improving generalizability, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to generalizability, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Strategy

14. Research Idea Generation 014: Diagnose Problem Selection for an ethics-aware deployment study

Objective: Build a high-clarity strategy for Problem Selection by improving reproducibility, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to reproducibility, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Execution

15. Research Idea Generation 015: Structure Feasibility Scan for a longitudinal dataset

Objective: Build a high-clarity strategy for Feasibility Scan by improving ethical compliance, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to ethical compliance, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Optimization

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16. Research Idea Generation 016: Interrogate Novelty Discovery for a high-impact conference submission

Objective: Build a high-clarity strategy for Novelty Discovery by improving cost efficiency, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to cost efficiency, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Planning

17. Research Idea Generation 017: Design Hypothesis Framing for an emerging technology review

Objective: Build a high-clarity strategy for Hypothesis Framing by improving computational efficiency, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to computational efficiency, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Methodology

18. Research Idea Generation 018: Refine Problem Selection for a policy-oriented manuscript

Objective: Build a high-clarity strategy for Problem Selection by improving publication readiness, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to publication readiness, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Validation

19. Research Idea Generation 019: Synthesize Feasibility Scan for an industry-backed experiment

Objective: Build a high-clarity strategy for Feasibility Scan by improving stakeholder clarity, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to stakeholder clarity, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Writing

20. Research Idea Generation 020: Translate Novelty Discovery for a multi-phase project

Objective: Build a high-clarity strategy for Novelty Discovery by improving novelty, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to novelty, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Reasoning

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21. Research Idea Generation 021: Model Hypothesis Framing for a method benchmarking campaign

Objective: Build a high-clarity strategy for Hypothesis Framing by improving feasibility, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to feasibility, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Strategy

22. Research Idea Generation 022: Reframe Problem Selection for a reproducibility-sensitive investigation

Objective: Build a high-clarity strategy for Problem Selection by improving rigor, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to rigor, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Execution

23. Research Idea Generation 023: Evaluate Feasibility Scan for a cross-disciplinary study

Objective: Build a high-clarity strategy for Feasibility Scan by improving generalizability, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to generalizability, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Optimization

24. Research Idea Generation 024: Validate Novelty Discovery for an ethics-aware deployment study

Objective: Build a high-clarity strategy for Novelty Discovery by improving reproducibility, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to reproducibility, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Planning

25. Research Idea Generation 025: Map Hypothesis Framing for a longitudinal dataset

Objective: Build a high-clarity strategy for Hypothesis Framing by improving ethical compliance, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to ethical compliance, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Methodology

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26. Research Idea Generation 026: Prioritize Problem Selection for a high-impact conference submission

Objective: Build a high-clarity strategy for Problem Selection by improving cost efficiency, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to cost efficiency, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Validation

27. Research Idea Generation 027: Forecast Feasibility Scan for an emerging technology review

Objective: Build a high-clarity strategy for Feasibility Scan by improving computational efficiency, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to computational efficiency, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Writing

28. Research Idea Generation 028: Operationalize Novelty Discovery for a policy-oriented manuscript

Objective: Build a high-clarity strategy for Novelty Discovery by improving publication readiness, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to publication readiness, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Reasoning

29. Research Idea Generation 029: Compare Hypothesis Framing for an industry-backed experiment

Objective: Build a high-clarity strategy for Hypothesis Framing by improving stakeholder clarity, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to stakeholder clarity, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Strategy

30. Research Idea Generation 030: Optimize Problem Selection for a multi-phase project

Objective: Build a high-clarity strategy for Problem Selection by improving novelty, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to novelty, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Execution

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31. Research Idea Generation 031: Diagnose Feasibility Scan for a method benchmarking campaign

Objective: Build a high-clarity strategy for Feasibility Scan by improving feasibility, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to feasibility, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Optimization

32. Research Idea Generation 032: Structure Novelty Discovery for a reproducibility-sensitive investigation

Objective: Build a high-clarity strategy for Novelty Discovery by improving rigor, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to rigor, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Planning

33. Research Idea Generation 033: Interrogate Hypothesis Framing for a cross-disciplinary study

Objective: Build a high-clarity strategy for Hypothesis Framing by improving generalizability, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to generalizability, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Methodology

34. Research Idea Generation 034: Design Problem Selection for an ethics-aware deployment study

Objective: Build a high-clarity strategy for Problem Selection by improving reproducibility, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to reproducibility, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Validation

35. Research Idea Generation 035: Refine Feasibility Scan for a longitudinal dataset

Objective: Build a high-clarity strategy for Feasibility Scan by improving ethical compliance, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to ethical compliance, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Writing

36. Research Idea Generation 036: Synthesize Novelty Discovery for a high-impact conference submission

Objective: Build a high-clarity strategy for Novelty Discovery by improving cost efficiency, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to cost efficiency, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Reasoning

37. Research Idea Generation 037: Translate Hypothesis Framing for an emerging technology review

Objective: Build a high-clarity strategy for Hypothesis Framing by improving computational efficiency, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to computational efficiency, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Strategy

38. Research Idea Generation 038: Model Problem Selection for a policy-oriented manuscript

Objective: Build a high-clarity strategy for Problem Selection by improving publication readiness, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to publication readiness, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Execution

39. Research Idea Generation 039: Reframe Feasibility Scan for an industry-backed experiment

Objective: Build a high-clarity strategy for Feasibility Scan by improving stakeholder clarity, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to stakeholder clarity, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Optimization

40. Research Idea Generation 040: Evaluate Novelty Discovery for a multi-phase project

Objective: Build a high-clarity strategy for Novelty Discovery by improving novelty, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to novelty, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Planning

41. Research Idea Generation 041: Validate Hypothesis Framing for a method benchmarking campaign

Objective: Build a high-clarity strategy for Hypothesis Framing by improving feasibility, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to feasibility, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Methodology

42. Research Idea Generation 042: Map Problem Selection for a reproducibility-sensitive investigation

Objective: Build a high-clarity strategy for Problem Selection by improving rigor, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to rigor, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Validation

43. Research Idea Generation 043: Prioritize Feasibility Scan for a cross-disciplinary study

Objective: Build a high-clarity strategy for Feasibility Scan by improving generalizability, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to generalizability, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Writing

44. Research Idea Generation 044: Forecast Novelty Discovery for an ethics-aware deployment study

Objective: Build a high-clarity strategy for Novelty Discovery by improving reproducibility, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to reproducibility, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Reasoning

45. Research Idea Generation 045: Operationalize Hypothesis Framing for a longitudinal dataset

Objective: Build a high-clarity strategy for Hypothesis Framing by improving ethical compliance, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to ethical compliance, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Strategy

46. Research Idea Generation 046: Compare Problem Selection for a high-impact conference submission

Objective: Build a high-clarity strategy for Problem Selection by improving cost efficiency, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to cost efficiency, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Execution

47. Research Idea Generation 047: Optimize Feasibility Scan for an emerging technology review

Objective: Build a high-clarity strategy for Feasibility Scan by improving computational efficiency, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to computational efficiency, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Optimization

48. Research Idea Generation 048: Diagnose Novelty Discovery for a policy-oriented manuscript

Objective: Build a high-clarity strategy for Novelty Discovery by improving publication readiness, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to publication readiness, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Planning

49. Research Idea Generation 049: Structure Hypothesis Framing for an industry-backed experiment

Objective: Build a high-clarity strategy for Hypothesis Framing by improving stakeholder clarity, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to stakeholder clarity, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Methodology

50. Research Idea Generation 050: Interrogate Problem Selection for a multi-phase project

Objective: Build a high-clarity strategy for Problem Selection by improving novelty, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to novelty, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Validation

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51. Research Idea Generation 051: Design Feasibility Scan for a method benchmarking campaign

Objective: Build a high-clarity strategy for Feasibility Scan by improving feasibility, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to feasibility, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Writing

52. Research Idea Generation 052: Refine Novelty Discovery for a reproducibility-sensitive investigation

Objective: Build a high-clarity strategy for Novelty Discovery by improving rigor, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to rigor, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Reasoning

53. Research Idea Generation 053: Synthesize Hypothesis Framing for a cross-disciplinary study

Objective: Build a high-clarity strategy for Hypothesis Framing by improving generalizability, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to generalizability, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Strategy

54. Research Idea Generation 054: Translate Problem Selection for an ethics-aware deployment study

Objective: Build a high-clarity strategy for Problem Selection by improving reproducibility, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to reproducibility, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Execution

55. Research Idea Generation 055: Model Feasibility Scan for a longitudinal dataset

Objective: Build a high-clarity strategy for Feasibility Scan by improving ethical compliance, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to ethical compliance, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Optimization

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56. Research Idea Generation 056: Reframe Novelty Discovery for a high-impact conference submission

Objective: Build a high-clarity strategy for Novelty Discovery by improving cost efficiency, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to cost efficiency, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Planning

57. Research Idea Generation 057: Evaluate Hypothesis Framing for an emerging technology review

Objective: Build a high-clarity strategy for Hypothesis Framing by improving computational efficiency, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to computational efficiency, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Methodology

58. Research Idea Generation 058: Validate Problem Selection for a policy-oriented manuscript

Objective: Build a high-clarity strategy for Problem Selection by improving publication readiness, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to publication readiness, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Validation

59. Research Idea Generation 059: Map Feasibility Scan for an industry-backed experiment

Objective: Build a high-clarity strategy for Feasibility Scan by improving stakeholder clarity, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to stakeholder clarity, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Writing

60. Research Idea Generation 060: Prioritize Novelty Discovery for a multi-phase project

Objective: Build a high-clarity strategy for Novelty Discovery by improving novelty, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to novelty, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Reasoning

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61. Research Idea Generation 061: Forecast Hypothesis Framing for a method benchmarking campaign

Objective: Build a high-clarity strategy for Hypothesis Framing by improving feasibility, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to feasibility, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Strategy

62. Research Idea Generation 062: Operationalize Problem Selection for a reproducibility-sensitive investigation

Objective: Build a high-clarity strategy for Problem Selection by improving rigor, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to rigor, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Execution

63. Research Idea Generation 063: Compare Feasibility Scan for a cross-disciplinary study

Objective: Build a high-clarity strategy for Feasibility Scan by improving generalizability, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to generalizability, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Optimization

64. Research Idea Generation 064: Optimize Novelty Discovery for an ethics-aware deployment study

Objective: Build a high-clarity strategy for Novelty Discovery by improving reproducibility, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to reproducibility, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Planning

65. Research Idea Generation 065: Diagnose Hypothesis Framing for a longitudinal dataset

Objective: Build a high-clarity strategy for Hypothesis Framing by improving ethical compliance, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to ethical compliance, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Methodology

66. Research Idea Generation 066: Structure Problem Selection for a high-impact conference submission

Objective: Build a high-clarity strategy for Problem Selection by improving cost efficiency, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to cost efficiency, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Validation

67. Research Idea Generation 067: Interrogate Feasibility Scan for an emerging technology review

Objective: Build a high-clarity strategy for Feasibility Scan by improving computational efficiency, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to computational efficiency, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Writing

68. Research Idea Generation 068: Design Novelty Discovery for a policy-oriented manuscript

Objective: Build a high-clarity strategy for Novelty Discovery by improving publication readiness, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to publication readiness, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Reasoning

69. Research Idea Generation 069: Refine Hypothesis Framing for an industry-backed experiment

Objective: Build a high-clarity strategy for Hypothesis Framing by improving stakeholder clarity, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to stakeholder clarity, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Strategy

70. Research Idea Generation 070: Synthesize Problem Selection for a multi-phase project

Objective: Build a high-clarity strategy for Problem Selection by improving novelty, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to novelty, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Execution

71. Research Idea Generation 071: Translate Feasibility Scan for a method benchmarking campaign

Objective: Build a high-clarity strategy for Feasibility Scan by improving feasibility, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to feasibility, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Optimization

72. Research Idea Generation 072: Model Novelty Discovery for a reproducibility-sensitive investigation

Objective: Build a high-clarity strategy for Novelty Discovery by improving rigor, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to rigor, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Planning

73. Research Idea Generation 073: Reframe Hypothesis Framing for a cross-disciplinary study

Objective: Build a high-clarity strategy for Hypothesis Framing by improving generalizability, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to generalizability, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Methodology

74. Research Idea Generation 074: Evaluate Problem Selection for an ethics-aware deployment study

Objective: Build a high-clarity strategy for Problem Selection by improving reproducibility, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to reproducibility, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Validation

75. Research Idea Generation 075: Validate Feasibility Scan for a longitudinal dataset

Objective: Build a high-clarity strategy for Feasibility Scan by improving ethical compliance, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to ethical compliance, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Writing

76. Research Idea Generation 076: Map Novelty Discovery for a high-impact conference submission

Objective: Build a high-clarity strategy for Novelty Discovery by improving cost efficiency, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to cost efficiency, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Reasoning

77. Research Idea Generation 077: Prioritize Hypothesis Framing for an emerging technology review

Objective: Build a high-clarity strategy for Hypothesis Framing by improving computational efficiency, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to computational efficiency, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Strategy

78. Research Idea Generation 078: Forecast Problem Selection for a policy-oriented manuscript

Objective: Build a high-clarity strategy for Problem Selection by improving publication readiness, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to publication readiness, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Execution

79. Research Idea Generation 079: Operationalize Feasibility Scan for an industry-backed experiment

Objective: Build a high-clarity strategy for Feasibility Scan by improving stakeholder clarity, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to stakeholder clarity, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Optimization

80. Research Idea Generation 080: Compare Novelty Discovery for a multi-phase project

Objective: Build a high-clarity strategy for Novelty Discovery by improving novelty, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to novelty, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Planning

81. Research Idea Generation 081: Optimize Hypothesis Framing for a method benchmarking campaign

Objective: Build a high-clarity strategy for Hypothesis Framing by improving feasibility, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to feasibility, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Methodology

82. Research Idea Generation 082: Diagnose Problem Selection for a reproducibility-sensitive investigation

Objective: Build a high-clarity strategy for Problem Selection by improving rigor, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to rigor, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Validation

83. Research Idea Generation 083: Structure Feasibility Scan for a cross-disciplinary study

Objective: Build a high-clarity strategy for Feasibility Scan by improving generalizability, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to generalizability, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Writing

84. Research Idea Generation 084: Interrogate Novelty Discovery for an ethics-aware deployment study

Objective: Build a high-clarity strategy for Novelty Discovery by improving reproducibility, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to reproducibility, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Reasoning

85. Research Idea Generation 085: Design Hypothesis Framing for a longitudinal dataset

Objective: Build a high-clarity strategy for Hypothesis Framing by improving ethical compliance, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to ethical compliance, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Strategy

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86. Research Idea Generation 086: Refine Problem Selection for a high-impact conference submission

Objective: Build a high-clarity strategy for Problem Selection by improving cost efficiency, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to cost efficiency, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Execution

87. Research Idea Generation 087: Synthesize Feasibility Scan for an emerging technology review

Objective: Build a high-clarity strategy for Feasibility Scan by improving computational efficiency, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to computational efficiency, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Optimization

88. Research Idea Generation 088: Translate Novelty Discovery for a policy-oriented manuscript

Objective: Build a high-clarity strategy for Novelty Discovery by improving publication readiness, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to publication readiness, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Planning

89. Research Idea Generation 089: Model Hypothesis Framing for an industry-backed experiment

Objective: Build a high-clarity strategy for Hypothesis Framing by improving stakeholder clarity, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to stakeholder clarity, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Methodology

90. Research Idea Generation 090: Reframe Problem Selection for a multi-phase project

Objective: Build a high-clarity strategy for Problem Selection by improving novelty, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to novelty, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Validation

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91. Research Idea Generation 091: Evaluate Feasibility Scan for a method benchmarking campaign

Objective: Build a high-clarity strategy for Feasibility Scan by improving feasibility, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to feasibility, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Writing

92. Research Idea Generation 092: Validate Novelty Discovery for a reproducibility-sensitive investigation

Objective: Build a high-clarity strategy for Novelty Discovery by improving rigor, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to rigor, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Reasoning

93. Research Idea Generation 093: Map Hypothesis Framing for a cross-disciplinary study

Objective: Build a high-clarity strategy for Hypothesis Framing by improving generalizability, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to generalizability, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Strategy

94. Research Idea Generation 094: Prioritize Problem Selection for an ethics-aware deployment study

Objective: Build a high-clarity strategy for Problem Selection by improving reproducibility, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to reproducibility, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Execution

95. Research Idea Generation 095: Forecast Feasibility Scan for a longitudinal dataset

Objective: Build a high-clarity strategy for Feasibility Scan by improving ethical compliance, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to ethical compliance, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Optimization

96. Research Idea Generation 096: Operationalize Novelty Discovery for a high-impact conference submission

Objective: Build a high-clarity strategy for Novelty Discovery by improving cost efficiency, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to cost efficiency, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Planning

97. Research Idea Generation 097: Compare Hypothesis Framing for an emerging technology review

Objective: Build a high-clarity strategy for Hypothesis Framing by improving computational efficiency, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to computational efficiency, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Methodology

98. Research Idea Generation 098: Optimize Problem Selection for a policy-oriented manuscript

Objective: Build a high-clarity strategy for Problem Selection by improving publication readiness, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to publication readiness, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Validation

99. Research Idea Generation 099: Diagnose Feasibility Scan for an industry-backed experiment

Objective: Build a high-clarity strategy for Feasibility Scan by improving stakeholder clarity, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to stakeholder clarity, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Writing

100. Research Idea Generation 100: Structure Novelty Discovery for a multi-phase project

Objective: Build a high-clarity strategy for Novelty Discovery by improving novelty, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to novelty, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Reasoning

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101. Research Idea Generation 101: Interrogate Hypothesis Framing for a method benchmarking campaign

Objective: Build a high-clarity strategy for Hypothesis Framing by improving feasibility, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to feasibility, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Strategy

102. Research Idea Generation 102: Design Problem Selection for a reproducibility-sensitive investigation

Objective: Build a high-clarity strategy for Problem Selection by improving rigor, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to rigor, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Execution

103. Research Idea Generation 103: Refine Feasibility Scan for a cross-disciplinary study

Objective: Build a high-clarity strategy for Feasibility Scan by improving generalizability, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to generalizability, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Optimization

104. Research Idea Generation 104: Synthesize Novelty Discovery for an ethics-aware deployment study

Objective: Build a high-clarity strategy for Novelty Discovery by improving reproducibility, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to reproducibility, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Planning

105. Research Idea Generation 105: Translate Hypothesis Framing for a longitudinal dataset

Objective: Build a high-clarity strategy for Hypothesis Framing by improving ethical compliance, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to ethical compliance, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Methodology

106. Research Idea Generation 106: Model Problem Selection for a high-impact conference submission

Objective: Build a high-clarity strategy for Problem Selection by improving cost efficiency, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to cost efficiency, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Validation

107. Research Idea Generation 107: Reframe Feasibility Scan for an emerging technology review

Objective: Build a high-clarity strategy for Feasibility Scan by improving computational efficiency, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to computational efficiency, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Writing

108. Research Idea Generation 108: Evaluate Novelty Discovery for a policy-oriented manuscript

Objective: Build a high-clarity strategy for Novelty Discovery by improving publication readiness, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to publication readiness, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Reasoning

109. Research Idea Generation 109: Validate Hypothesis Framing for an industry-backed experiment

Objective: Build a high-clarity strategy for Hypothesis Framing by improving stakeholder clarity, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to stakeholder clarity, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Strategy

110. Research Idea Generation 110: Map Problem Selection for a multi-phase project

Objective: Build a high-clarity strategy for Problem Selection by improving novelty, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to novelty, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Execution

111. Research Idea Generation 111: Prioritize Feasibility Scan for a method benchmarking campaign

Objective: Build a high-clarity strategy for Feasibility Scan by improving feasibility, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to feasibility, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Optimization

112. Research Idea Generation 112: Forecast Novelty Discovery for a reproducibility-sensitive investigation

Objective: Build a high-clarity strategy for Novelty Discovery by improving rigor, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to rigor, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Planning

113. Research Idea Generation 113: Operationalize Hypothesis Framing for a cross-disciplinary study

Objective: Build a high-clarity strategy for Hypothesis Framing by improving generalizability, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to generalizability, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Methodology

114. Research Idea Generation 114: Compare Problem Selection for an ethics-aware deployment study

Objective: Build a high-clarity strategy for Problem Selection by improving reproducibility, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to reproducibility, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Validation

115. Research Idea Generation 115: Optimize Feasibility Scan for a longitudinal dataset

Objective: Build a high-clarity strategy for Feasibility Scan by improving ethical compliance, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to ethical compliance, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Writing

116. Research Idea Generation 116: Diagnose Novelty Discovery for a high-impact conference submission

Objective: Build a high-clarity strategy for Novelty Discovery by improving cost efficiency, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to cost efficiency, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Reasoning

117. Research Idea Generation 117: Structure Hypothesis Framing for an emerging technology review

Objective: Build a high-clarity strategy for Hypothesis Framing by improving computational efficiency, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Hypothesis Framing. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to computational efficiency, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Hypothesis Framing.

Difficulty: Beginner

Category: Strategy

118. Research Idea Generation 118: Interrogate Problem Selection for a policy-oriented manuscript

Objective: Build a high-clarity strategy for Problem Selection by improving publication readiness, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Problem Selection. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to publication readiness, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and hypothesis matrix with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade hypothesis matrix containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Problem Selection.

Difficulty: Intermediate

Category: Execution

119. Research Idea Generation 119: Design Feasibility Scan for an industry-backed experiment

Objective: Build a high-clarity strategy for Feasibility Scan by improving stakeholder clarity, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Feasibility Scan. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to stakeholder clarity, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and priority map with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade priority map containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Feasibility Scan.

Difficulty: Advanced

Category: Optimization

120. Research Idea Generation 120: Refine Novelty Discovery for a multi-phase project

Objective: Build a high-clarity strategy for Novelty Discovery by improving novelty, reducing ambiguity, and aligning decisions with research objectives.

Prompt: You are an expert research strategist. Analyze the project context below and produce a structured plan focused on Novelty Discovery. Context: discipline={discipline}, dataset={dataset}, time horizon={timeline}, resource limits={resources}. Deliver: (1) assumptions, (2) stepwise method, (3) evaluation criteria linked to novelty, (4) risk controls, and (5) an implementation schedule. Conclude with measurable success indicators and a concise quality checklist.

Example: Example input: discipline=computational biology; dataset=single-cell RNA-seq cohort; timeline=12 weeks; resources=2 researchers and 1 GPU node. Example outcome: a staged protocol and research roadmap with milestones.

Tips: State assumptions explicitly; define boundaries before proposing methods; tie every recommendation to a measurable criterion; include one contingency branch for likely failure points.

Expected Output: A publication-grade research roadmap containing rationale, method choices, decision criteria, risk mitigations, and implementation priorities for Novelty Discovery.

Difficulty: Expert

Category: Planning

Summary

Applying these 120 prompts systematically improves decision quality, writing clarity, and publication readiness across research cycles.